

Deepak Mallampati

APPLIED AI ENGINEER - FORWARD DEPLOYED / FIELD ENGINEERING

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US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years shipping production-grade LLM inference, fine-tuning, and agentic systems across enterprise and regulated domains. Delivered **42% token reduction** in agentic orchestration, **p95 latency from 1.5s to 800ms** at 100K+ daily requests, and QLoRA fine-tuning of a 27B model on H100 for controllable long-form generation. Combines hands-on inference cost/latency engineering, model selection (GPT-4o/Claude/Gemini eval, +27% task success), and cross-team delivery (25+ AI agents across 20+ teams) - exactly the build-and-deploy profile partner customers need at the Azure AI Foundry boundary.

CORE COMPETENCIES

LLM Inference Optimization • Fine-tuning (LoRA, QLoRA, PEFT) • Agentic Orchestration • RAG / Retrieval • Model Selection & Evals • Production Hardening • Kubernetes & Cloud Infra • Cross-team Delivery

LLMs & Inference: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA, Qwen | LangChain/LangGraph | Prompt compression, semantic caching, model routing | Hugging Face Transformers

Fine-tuning: LoRA, QLoRA (4-bit on H100), PEFT | Instruction-tuning datasets | Schema-conditioned generation

Eval & Observability: RAGAS, BERTScore, custom domain metrics | Latency & accuracy dashboards | Content safety controls

Cloud & Infra: Azure (OpenAI, AI Services, Content Safety), AWS (Bedrock, SageMaker, Lambda), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps)

Vector / Retrieval: FAISS, Pinecone, Weaviate | Cross-encoder re-ranking | Hybrid search

Languages: Python, TypeScript/JavaScript, Java, SQL | FastAPI, React

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII redaction | Prompt injection mitigation

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing **token consumption by 42%** while sustaining task accuracy across multi-step workflows.
- Optimized inference cost and latency through **prompt compression, semantic caching, and model routing**, sustaining SLA targets while reducing per-query overhead - directly applicable to partner-side cost optimization.
- Engineered high-performance **RAG pipelines** over large healthcare corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) paired with latency/accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Built fault-tolerant automation infrastructure with scheduled orchestration and exponential-backoff retries, reducing failed production runs by **60%**.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed and shipped **25+ AI agents and workflows** across Vanguard's internal AI platform, partnered with 20+ teams across product, platform, and data engineering - the same cross-team delivery shape required for partner co-sell motions.
- Built **12 APIs and microservices** powering new AI capabilities, reducing **p95 latency from 1.5s to 800ms** and supporting **100,000+ requests daily**.
- Evaluated **GPT-4o, Claude Sonnet, and Gemini** on internal benchmarks, improving task success rates by **27%** - direct experience advising customers on model selection (SFT/DPO/RFT trade-offs).
- Engineered content safety controls and policy-compliant guardrails, reducing unsafe outputs by **42%** while maintaining response quality.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Fine-tuned **Qwen3-27B** via **4-bit QLoRA** on an **NVIDIA H100** using a curated 361-example instruction dataset, conditioning the model on a six-stage narrative schema for controllable long-form generation - hands-on fine-tuning artifact directly relevant to guiding partners on LoRA/RFT strategy.
- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, Laplace differential privacy) reducing re-identification risk from 3.38% → 0.32% while retaining 99% of baseline predictive performance.
- Authored a peer-review-ready IEEE-format conference paper documenting methodology, experiments, and results.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Developed supervised regression and classification models on operational pipeline data - equipment-failure prediction, anomaly detection, capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.

PROJECTS

Story Director LLM

A fine-tuned 27B language model (QLoRA on H100) that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline. End-to-end fine-tuning artifact - direct evidence of LoRA/PEFT delivery from data prep through hosted inference.

MangaLens

Shipped Chrome extension (live on Chrome Web Store) delivering real-time AI translation across 29+ manga/webtoon sites - production LLM application with measurable user impact.

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight - an end-to-end agentic system shipped as open source.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy on clinical readmission data - quantifies the privacy/utility trade-off across six anonymization configurations.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA: 3.70 / 4.0